

Software Requirements Specification (SRS)

Case Study 8 – SDG 7: EV Charging Station Management System

1. Introduction

1.1 Background of the Problem

The rapid adoption of Electric Vehicles (EVs) is transforming the transportation industry by reducing dependence on fossil fuels and lowering carbon emissions. However, the growth of EV usage has created significant pressure on the existing charging infrastructure. Many EV owners face difficulties locating nearby charging stations, checking charger availability, and waiting in long queues before charging their vehicles. Charging station operators also struggle to manage reservations, monitor equipment utilization, and provide efficient services to customers.

Most existing charging systems lack real-time monitoring, online reservation facilities, and centralized management. This results in inefficient utilization of charging stations and poor user experience. A digital EV Charging Station Management System can address these issues by providing real-time availability, online booking, secure payment, and efficient station management.

1.2 Importance of Solving the Problem

Developing an intelligent EV Charging Station Management System provides several benefits:

- Reduces waiting time for EV users.
 - Enables efficient utilization of charging stations.
 - Supports cashless digital payments.
 - Improves customer satisfaction.
 - Reduces traffic congestion around charging stations.
 - Encourages the adoption of electric vehicles.
 - Supports environmental sustainability by promoting clean energy transportation.
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1.3 Related SDG Goal(s)

SDG 7 – Affordable and Clean Energy

- Promote access to reliable and sustainable energy.
- Encourage the adoption of clean transportation.
- Improve energy efficiency through smart charging infrastructure.

Related Goals

- **SDG 9:** Industry, Innovation and Infrastructure
 - **SDG 11:** Sustainable Cities and Communities
 - **SDG 13:** Climate Action
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1.4 Objectives of the Case Study

The objectives of this project are:

- Develop a centralized EV charging management platform.
 - Provide real-time charger availability.
 - Enable advance slot booking.
 - Reduce charging waiting time.
 - Improve charging station utilization.
 - Support secure online payments.
 - Monitor charging sessions efficiently.
 - Generate analytical reports for operators.
 - Promote sustainable transportation.
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2. Problem Understanding

2.1 Problem Description

As the number of electric vehicles increases, the demand for charging stations also rises. EV users often spend significant time searching for available charging stations and waiting in long queues. Most charging stations do not provide real-time availability information or reservation facilities, causing inconvenience to users.

Charging station operators also face challenges in monitoring charger utilization, scheduling maintenance, and managing customer reservations. Without a centralized system, resource utilization becomes inefficient.

A smart software platform can solve these challenges by enabling users to locate nearby charging stations, reserve charging slots, make online payments, and receive notifications. Operators can monitor charging sessions, manage equipment, and analyze usage statistics.

2.2 Objectives

The proposed software aims to:

- Reduce manual booking processes.
 - Improve charging station utilization.
 - Reduce waiting time.
 - Increase operational efficiency.
 - Improve transparency.
 - Provide real-time monitoring.
 - Enhance user experience.
 - Support digital payments.
 - Generate reports.
 - Improve customer satisfaction.
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2.3 Scope

In Scope

- User Registration
- Login Authentication
- Charging Station Search
- Real-time Availability
- Online Reservation
- Online Payment
- Notifications
- Charging History
- Admin Dashboard
- Operator Dashboard
- Reports

Out of Scope

- EV manufacturing
- Vehicle maintenance
- Battery replacement services

- Physical charger installation
- Hardware development
- Electricity generation
- Third-party fleet management

3. Stakeholder Analysis

Stakeholder	Role	Responsibilities	Challenges
Administrator	System Manager	Manage users, stations, reports	System maintenance
EV Customer	End User	Search, book, pay, charge	Waiting time
Charging Station Operator	Station Manager	Monitor chargers, reservations	Resource management
Government Agency	Regulatory Authority	Policy compliance	Infrastructure expansion
Payment Gateway	Payment Processing	Secure transactions	Transaction failures
Technician	Maintenance Staff	Repair chargers	Equipment failures
Support Staff	Customer Service	Handle complaints	Customer satisfaction

4. Existing System Analysis

4.1 Existing Process

Currently, many EV charging stations operate independently without centralized management. Users manually search for charging stations using maps or mobile applications. Most stations do not provide real-time availability or reservation facilities. Customers often arrive only to find chargers occupied, resulting in long waiting times. Operators manually manage charging sessions and maintenance schedules, leading to inefficient resource utilization.

4.2 Strengths

- Simple charging process.
- Available in many cities.
- Supports fast charging.
- Existing payment options.
- Basic mobile applications.

4.3 Weaknesses

- No real-time availability.
- Long waiting queues.
- Manual reservation process.
- Poor charger utilization.
- Limited monitoring.
- Lack of analytics.
- No centralized management.
- Inconsistent user experience.

4.4 Gap Analysis

Current systems fail to provide:

- Live charger status.
- Online reservation.
- Queue management.
- Smart notifications.
- Usage analytics.
- Predictive maintenance.
- Centralized monitoring.
- Efficient resource allocation.

5. Proposed Software Solution

5.1 Proposed System

The proposed Smart EV Charging Station Management System provides a centralized platform where users can locate nearby charging stations, check charger availability in real time, reserve charging slots, make secure online payments, and receive notifications.

Charging station operators can monitor charger utilization, manage reservations, and generate reports through an administrative dashboard.

5.2 Functional Modules

- User Registration

- Login
 - User Dashboard
 - Station Search
 - Real-time Availability
 - Reservation
 - Payment
 - Notifications
 - Charging History
 - Admin Panel
 - Operator Dashboard
 - Reports
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5.3 Functional Requirements

FR1: The system shall allow users to register.

FR2: The system shall authenticate users securely.

FR3: The system shall display nearby charging stations.

FR4: The system shall show real-time charger availability.

FR5: The system shall allow users to reserve charging slots.

FR6: The system shall support online payments.

FR7: The system shall send booking confirmations.

FR8: The system shall maintain charging history.

FR9: The system shall generate utilization reports.

FR10: The administrator shall manage users and charging stations.

5.4 Non-Functional Requirements

- High Performance
- Security
- Scalability
- Reliability
- Maintainability
- Availability

- Usability
 - Portability
 - Backup and Recovery
 - Fast Response Time
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5.5 Software Engineering Principles Applied

Modularity

Separate modules for users, operators, payments, and reports.

Scalability

Supports increasing numbers of users and charging stations.

Reliability

Ensures uninterrupted charging services.

Security

Uses encryption, authentication, and secure payments.

Maintainability

Easy software updates and bug fixes.

Reusability

Reusable software components.

Usability

Simple and user-friendly interface.

Flexibility

Supports future feature additions and integrations.

6. System Design

Include the following diagrams (create using Draw.io, Lucidchart, Figma, or Canva):

1. Use Case Diagram

- Actors: EV User, Operator, Administrator
- Use Cases: Register, Login, Search Station, Book Slot, Payment, Monitor Chargers, Generate Reports

2. Use Case Description (optional)

- Brief description of each use case, actors involved, preconditions, and outcomes.

3. Activity Diagram

- User logs in → Searches station → Checks availability → Books slot → Makes payment → Charges vehicle → Receives notification.

4. System Architecture Diagram

- User App/Web → API Server → Authentication → Database → Payment Gateway → Notification Service → Charging Station IoT/Monitoring.

5. Database ER Diagram

- Entities: User, ChargingStation, Charger, Reservation, Payment, Notification, Operator.

6. Wireframe / UI Screens

- Login Page
- Registration Page
- Dashboard
- Station Search Map
- Booking Screen
- Payment Page
- Admin Dashboard

7. Workflow Diagram

- Registration → Login → Search Station → Reserve → Pay → Charge → Feedback/History.

7. Methodology / SDLC Selection

Selected Model: Agile Scrum

Why Agile Scrum?

Agile Scrum is suitable because requirements may evolve as user feedback is gathered. Development can proceed in short iterations, enabling frequent testing and continuous improvement.

Advantages

- Flexible to changing requirements.
- Faster delivery of working software.
- Continuous customer feedback.
- Early bug detection.
- Improved team collaboration.

Suitability

Ideal for web and mobile applications that require regular feature updates and user-centric improvements.

Development Stages

1. Requirement Gathering
2. Sprint Planning
3. Design
4. Development
5. Testing
6. Review
7. Deployment
8. Maintenance

8. Implementation Plan

Phase	Duration
Requirement Gathering	1 Week
System Design	1 Week
Development	3 Weeks
Testing	1 Week
Deployment	1 Week
Maintenance	Ongoing

9. Expected Outcomes

Business Impact

- Improved charging station utilization.
- Increased operational efficiency.
- Higher revenue through optimized bookings.
- Better resource management.

User Benefits

- Reduced waiting time.
- Easy reservation process.

- Secure online payments.
- Real-time charger availability.
- Better overall charging experience.

Social Impact

- Encourages adoption of electric vehicles.
- Reduces carbon emissions.
- Supports sustainable transportation.
- Promotes digital services.

SDG Contribution

- Advances **SDG 7** by enabling efficient use of clean energy infrastructure.
 - Supports **SDG 9** through innovative digital infrastructure.
 - Contributes to **SDG 11** by improving sustainable urban mobility.
 - Helps achieve **SDG 13** by reducing greenhouse gas emissions.
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10. Risks and Limitations

- Dependence on reliable internet connectivity.
 - Data privacy concerns.
 - Cybersecurity threats.
 - Budget constraints for deployment and maintenance.
 - User adoption and training challenges.
 - Integration issues with different charging station providers.
 - Hardware failures or charger downtime.
 - Scalability challenges as the network grows.
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11. Future Enhancements

- AI-based charging demand prediction.
- Chatbot for customer support.
- Predictive maintenance using IoT sensors.
- Mobile application for Android and iOS.
- Blockchain-based secure energy transactions.
- Cloud-native deployment for scalability.

- Dynamic pricing based on demand.
 - Integration with renewable energy sources and smart grids.
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12. Conclusion

The proposed **EV Charging Station Management System** addresses key challenges faced by electric vehicle users and charging station operators by providing real-time charger availability, online reservations, secure payments, and centralized management. The solution improves operational efficiency, enhances user satisfaction, and supports sustainable transportation. By promoting efficient use of charging infrastructure and encouraging EV adoption, the system contributes significantly to **SDG 7: Affordable and Clean Energy**, while also supporting broader sustainability and smart city goals.

13. References

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